CONDITIONAL PROBABILITY AND INDEPENDENT EVENTS **14/11/25**

1. Sick people in an under-resourced hospital are each given a pill which is either a placebo or a homeopathic preparation. The probability of being given a homeopathic pill is 0.5, the probability of recovery is 0.4 and the probability of being given a homeopathic pill and recovering is 0.2. Show that recovery is independent of homeopathic treatment.

2. In a class of 30 boys, 11 own a bike only, 6 own a motorbike only, and 4 have both.
   (a) If a student is picked at random from this class, what is the probability that he has:
      (i) neither a bike nor a motorbike?
      (ii) a bike, but no motorbike?
   (b) Find the probability that a randomly picked student owns a bike, given that he owns a motorbike.
   (c) Are the events ‘owning a bike’ and ‘owning a motorbike’ independent?

3. A motorist's journey home involves passing through three sets of traffic lights. The motorist models the chances of being delayed at each set of lights as independent, and with probabilities 0.2, 0.6, and 0.5. Find, according to this model, the probability that the motorist:
   a. is delayed at all three sets of lights
   b. is delayed at exactly one set of lights
   c. is delayed at the second set of lights, given that she is delayed at exactly one set of lights.

4. Two events G and H are such that $P(G) = 0.4$, $P(H) = 0.25$ and $P(G \cup H) = 0.55$. Show that G and G are independent.

5. A card is drawn from a pack. Let *A*, *B* and *C* be the events ‘the card is a spade’, ‘the card is a court card’ and ‘the card is black’. Which of the pairs *A* and *B*, *A* and *C* and *B* and *C* are independent?

6. A and B are independent. $P(A') = \frac{5}{8}$, and $P(B') = \frac{5}{9}$. Find P(A∩B).

7. Two people set out independently to beat a particular computer game, their probabilities of success being $\frac{2}{3}$ and $\frac{1}{4}$ respectively. Find the probability that at least one of them will succeed.

8. A and B are two events such that $\mathrm{P}(A) = \frac{1}{4}$, $\mathrm{P}(B) = \frac{1}{3}$, and $\mathrm{P}(A \cup B) = \frac{1}{2}$. Find $\mathrm{P}(A \cap B)$.

9. Let *A* and *B* be events with P(A) = $\frac{1}{2}$, $P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{4}$. Find
   a. $\mathrm{P}(A \mid B)$
   b. $\mathrm{P}(B \mid A)$
   c. $\mathrm{P}(A \cup B)$
   d. $\mathrm{P}(A^C \cap B^C)$
   e. $\mathrm{P}(A^C \mid B^C)$
   f. $\mathrm{P}(B^C \mid A^C)$